PAPER_164 — High-Energy Dataset UQFF Validation Framework: CERN, GWOSC, EHT, Chandra

Session: *47 |* ***Date:*** *March 13, 2026 |* ***Thread:*** *7f9068 |* ***Domain:*** *§2.3*

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Abstract

This paper establishes the **High-Energy Dataset Validation Framework** for UQFF parameter calibration using four major multi-messenger datasets: CERN LHC (ATLAS 13 TeV + CMS 7 TeV), GWOSC O4a (including GW231123 225 M*sol merger), EHT Sgr A* 2017, and Chandra CSC 2.1. Each dataset constrains a specific UQFF MUGE term. A new observable—LHC collision energy* E*coll* = 13 TeV—enters the quantum uncertainty term as ΔE*vac, and the Osc*term becomes a variable parameter driven by GW O4 background amplitude.

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1. Dataset-to-UQFF Parameter Mapping

Dataset	Observable	Target UQFF Term	Updated Calibration
ATLAS 13 TeV, 65 TB	E_coll = 13 TeV	g*quantum (*Δ*E*vac)	ΔE_vac = 13 TeV/c^2
CMS 7 TeV	Cross-sections	ΔxΔp quantum bound	ΔxΔp ≥ ■/2 confirmed
GWOSC O4a + GW231123	GW background amplitude	Osc_term (variable)	Osc*term = h*GW·ω_GW^2
EHT Sgr A* (2017)	Shadow radius	a*aether*res	Re-calibrated a*aether*res
Chandra CSC 2.1	X-ray magnetar spectra	B/B_crit	B confirmed for SGR 1745
arXiv axion/ALP papers	ALP-photon coupling	a*Aether*freq	Dark energy coupling g_aγγ

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2. CERN LHC Quantum Uncertainty Calibration

2.1 Ecoll = 13 TeV → ΔEvac

From ATLAS Run 2 (2016–2018), E_coll = 13 TeV center-of-mass energy:

```
\Delta E_{vac} = \frac{E_{coll}}{V_{interaction}} =
\frac{13\,\text{TeV}}{(1\,\text{fm})^3}
```

```
= \frac{13 \times 1.602\times10^{-6}\,\text{J}}{(10^{-15})^3\,\text{m}^3} =
2.083\times10^{39}\,\text{J/m}^3
```

This updates the quantum term in PAPER_163 Function 6:

```
g_{quantum} = \frac{\hbar}{\Delta x \cdot \Delta p_{LHC}} \cdot \psi_{int} \cdot
\frac{2\pi}{t_H}
```

where $\Delta p_{LHC} = E_{coll}/(2c) = 6.5\,\text{TeV}/c$ for one beam.

2.2 Uncertainty Principle Bound Confirmation

$$\Delta x \cdot \Delta p = \frac{\hbar}{2}: \quad \Delta p = 6.5\,\text{TeV}/c \implies \Delta x = \frac{\hbar c}{2 \times 6.5\,\text{TeV}} \approx 1.5\times10^{-20}\,\text{m}$$

This is 10■ × smaller than the proton radius, confirming deep sub-nuclear UQFF coupling at collider energies.

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3. GWOSC O4a — Variable Osc_term

3.1 Previous Implementation

In PAPER*146 §2.2, Osc*term was set to a literal constant.

3.2 New Variable Osc_term

The O4a detector data (GWOSC, 2023-2024) provides:

- GW background strain: $h_{GW} \approx 2.5 \times 10^{-24}$ $Hz^{-1}/^{2}$ (stochastic background)
- Peak frequency: ω_GW ~ 2π × 100 Hz

$$\boxed{Osc_{term} = h_{GW} \cdot \omega_{GW}^2 \cdot r^2 \cdot \frac{M}{M_{BH,merger}}}$$

For GW231123 (225 M_sol):

$$Osc_{term}^{max} = 2.5\times10^{-24} \times (200\pi)^2 \times r^2 \times \frac{M}{225 M_\odot}$$

This makes Osc_term a **function of source distance** and merger mass, enabling correlated UQFF predictions for future GW events.

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4. EHT Sgr A* — a*aether*res Calibration

The Event Horizon Telescope (2017) resolved the Sgr A* shadow diameter:

- Observed shadow: 51.8 μas (theoretical for Kerr BH: 52±2 μas)
- Shadow radius: $r_{shadow} = (2.6 \pm 0.1) \times R_{Schwarzschild}$

This constrains the aether resonance term which contributes to the effective gravitational radius seen by photons:

$$a_{aether,res}^{EHT} = \frac{c^4}{G M_{SgrA^*}} \cdot \epsilon_{shadow}$$

where $\epsilon_{shadow} = (r_{obs} - r_{GR})/r_{GR} \approx 0.02$ (2% EHT precision limit).

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5. Chandra CSC — B/B_crit for Magnetars

Chandra soft X-ray spectra for SGR 1745-2900 (2013-2023 campaign):

- Confirmed B_surface = 2.3×10^{1}■ G = 2.3×10^{10} T
- B_crit (quantum) = 4.4×10^{13} T

```
B/B_{crit} = 2.3\times10^{10} / 4.4\times10^{13} = 5.23\times10^{-4}
```

→ MUGE suppression factor: `f_{super} = 1 - 5.23\times10^{-4} \approx 0.9995`

The magnetar is nearly unsuppressed → resonance MUGE dominates → consistent with PAPER*155 Newtonian emergence proof (*β ≈ *1 for this B/B*crit ratio).

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6. ALP Dark Energy Coupling → a*Aether*freq

Axion-Like Particle (ALP) photon coupling from arXiv papers (2023-2025):

- CAST/ABRACADABRA constraint: `g_{a\gamma\gamma} < 6\times10^{-11}` GeV^{-1}
- UQFF connection: a*Aether*freq represents the dark energy field resonance

```
a_{Aether,freq} = g_{a\gamma\gamma} \cdot \rho_{DM,local} \cdot c^2 \cdot \omega_{ALP}
```

where `\omega_{ALP} = m_{ALP}c^2/\hbar \sim 10^{-22}` to `10^{-12}` Hz (fuzzy dark matter range).

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7. Updated Parameter Table

Parameter	Old Value	New Value	Calibrating Dataset
Osc_term	const	$hGW \cdot \omega GW^2 \cdot r^2 \cdot M/m$	GWOSC O4a
ΔE_vac	undefined	13 TeV / $(1\ \mathrm{fm})^3$	ATLAS 13 TeV
a*aether*res	calibrated	±2% EHT constraint	EHT Sgr A* 2017
B/B_crit	2.3e10/4.4e13	same confirmed	Chandra CSC 2.1
a*Aether*freq	constant	$g a\gamma\gamma \cdot \rho DM \cdot c^2 \cdot \omega$_ALP	arXiv ALP surveys

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Status: ■ Complete | **CP Stage:** CP2/CP3 **Supersedes:** N/A (extends calibration) | **Related:** PAPER*064 (calibrated constants), PAPER*146 (Osc*term in 12-term), PAPER*167 (GW231123 event)
